



University
of Exeter

Text Mining & NLP

COMM040

Dr. Rodrigo Wilkens

Today

- By the end of this session, students will be able to:
 1. Analyse faithfulness, plausibility, and bias risks in explainability methods for NLP models.
 2. Explain feature attribution, gradient-based, and model-based explainability techniques in modern NLP.
 3. Interpret token-level importance using LIME, SHAP, saliency maps, and Integrated Gradients.
 4. Evaluate attention-based and probing approaches regarding encoding, usage, and causal faithfulness.

Outline

Foundations of Explainability in NLP

Explainability Techniques in NLP

Explainability in NLP: Why Now?

- Rapid progress in NLP and Large Language Models (LLMs)
 - Critical applications: Healthcare Finance Customer systems
 - High performance but low transparency
 - Decisions often cannot be understood by users
- Opacity $\Rightarrow$ Trust Accountability Debugging

Explainability in NLP (XNLP)

- XNLP = application of XAI techniques to language models
- Goes beyond prediction:
 - Answers **why**, **how**, and **what influenced** the output
- Supports:
 - Human understanding
 - Trust in model decisions
 - Auditing and validation

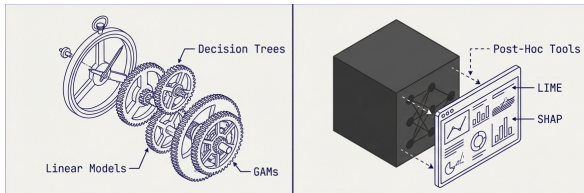
Interpretability vs Explainability

- **Interpretability**

- Model is inherently understandable
- Example: linear models, logistic regression

- **Explainability**

- External methods explain complex models
- Example: LIME, SHAP



Risks of Non-Explainable NLP

- Biases from training data may remain hidden
- Models may reinforce stereotypes or inequalities
- Errors are difficult to detect and correct
- Critical decisions may be unjustified

Use interpretable models

Local vs Global Explanations

- Types of Explanation:
 - **Local explanations**
 - ◇ Explain a single prediction
 - ◇ Example: why this sentence was classified as toxic
 - **Global explanations**
 - ◇ Describe overall model behavior
 - ◇ Example: which features influence predictions overall

Faithfulness vs Plausibility

- Types of Explanation:
 - **Faithfulness**
 - ◇ Explanation reflects the true model reasoning
 - **Plausibility**
 - ◇ Explanation is understandable to humans
 - These goals may conflict

Evolution of Interpretability in NLP



University
of Exeter

- Feature Importance in Early NLP
 - Words represented as features
 - Model coefficients indicate importance
 - Example:
 - ◇ Positive weight $\rightarrow$ increases prediction
 - ◇ Negative weight $\rightarrow$ decreases prediction

Evolution of Interpretability in NLP



University
of Exeter

- From Embeddings to Black-Box Models
 - Word embeddings:
 - ◇ Capture semantic similarity
 - Transformer models:
 - ◇ Capture complex dependencies
 - ◇ Difficult to interpret
 - ◇ Motivates the need for XAI methods

Outline

Foundations of Explainability in NLP

Explainability Techniques in NLP

Families of Explainability Methods



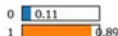
- Main categories of XAI methods in NLP:
 1. Feature attribution methods:
LIME and SHAP
 2. Gradient-based methods
 3. Model-based / architectural methods

1. Feature Attribution Methods

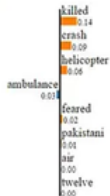
- LIME: Local Interpretable Model-Agnostic Explanations
 - Post-hoc, model-agnostic method
 - Explains **individual predictions**
 - Core idea:
 - ◇ Approximate the model locally with a simple model

1 = disaster class, 0 = non-disaster class

Prediction probabilities



0



1

Text with highlighted words

twelve feared **killed** pakistani air ambulance helicopter
crash

1. Feature Attribution Methods

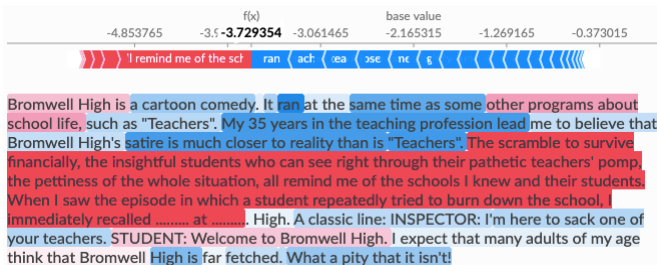
- How LIME Works (Step-by-step)
 - Given an input text:
 1. Generate perturbed versions (remove words)
 2. Get predictions for perturbed samples
 3. Fit a simple model (e.g., linear)
 4. Extract feature importance from this local model

Limitations of LIME

- Perturbed texts may be:
 - Linguistically unrealistic
- Ignores:
 - Syntax
 - Semantic dependencies
- Local explanations may be:
 - Unstable

1. Feature Attribution Methods

- SHAP: SHapley Additive exPlanations
 - Based on game theory (Shapley values)
 - Assigns contribution to each feature
 - Key idea:
 - ◇ Fair distribution of contribution across features



1. Feature Attribution Methods

- How SHAP Works
 - Considers:
 - ◇ All possible subsets of features
 - Measures:
 - ◇ Marginal contribution of each feature
 - Produces:
 - ◇ Additive explanation of prediction

1. Feature Attribution Methods

- SHAP in NLP
 - Words or tokens as features
 - Output:
 - ◇ Contribution of each word
 - Can provide:
 - ◇ Local explanations
 - ◇ Global explanations

1. Feature Attribution Methods

- Limitations of SHAP
 - Computationally expensive
 - Requires approximations (e.g., KernelSHAP)
 - Challenges in NLP:
 - ◇ Interpretation of fragmented tokens

2. Gradient-based Explanations

- Saliency Maps
 - Compute gradient of output with respect to input
 - Measure sensitivity:
 - ◇ How much does the output change if input changes?
 - Applied to NLP:
 - ◇ Gradients over word embeddings

2. Gradient-based Explanations

- Saliency Maps in NLP
 - Step-by-step:
 1. Input $\rightarrow$ tokens
 2. Tokens $\rightarrow$ embeddings
 3. Forward pass $\rightarrow$ output score y
 4. Backprop
 5. gradient norm

2. Gradient-based Explanations

- Advantages of Saliency Maps
 - Simple to compute
 - Computationally efficient
 - Directly tied to model internals
- Limitations of Saliency Maps
 - Sensitive to small variations in gradients
 - Hard to interpret for deep architectures
 - Issue:
 - ◇ Gradients may not reflect global importance

2. Gradient-based Explanations

- Integrated Gradients (IG)
 - Computes importance by integrating gradients
 - A baseline input (e.g., empty or **neutral text**)
 - ◇ Sequence of [PAD] tokens
 - ◇ Sequence of [UNK] tokens

2. Gradient-based Explanations (IG) University of Exeter

- step-by-step
 1. Input $\rightarrow$ tokens
 2. Tokens $\rightarrow$ embeddings
 3. Choose a baseline embedding
 4. Create interpolated embeddings (*cont*)
 5. Forward pass for each interpolated point
 6. Backprop for each interpolated point
 7. Average gradients along the path
 8. Multiply by (input-baseline) (*cont*)
 9. Aggregate per token into a scalar score.

IG (cont)

- Create interpolated embeddings between baseline and input
 - baseline = $[2, 4]$
 - input = $[20, 40]$

α	Operation	Result
0.0	$[2,4] + 0 \cdot [18,36]$	$[2, 4]$
0.25	$[2,4] + 0.25 \cdot [18,36]$	$[6.5, 13]$
0.5	$[2,4] + 0.5 \cdot [18,36]$	$[11, 22]$
0.75	$[2,4] + 0.75 \cdot [18,36]$	$[15.5, 31]$
1.0	$[2,4] + 1 \cdot [18,36]$	$[20, 40]$

IG (cont)

- Multiply by (input-baseline)
- Baseline = $[2, 4]$
- Input = $[20, 40]$
- Input - Baseline = $[18, 36]$
- Average Gradient = $[0.1, 0.2]$
- IG = $[18 \times 0.1, 36 \times 0.2] = [1.8, 7.2]$

2. Gradient-based Explanations

- Limitations of Integrated Gradients
 - Baseline choice influences results
 - More computationally expensive than saliency
 - Still not always intuitive for users

3. Model-based methods

- Layer-wise Relevance Propagation (LRP)
 - Explanation method for neural networks
 - Redistributes prediction relevance:
 - ◊ From output back to input
 - Goal:
 - ◊ Identify contribution of each input feature

3. Model-based methods

- Core Idea of LRP
 - Start from model output (prediction score)
 - Propagate relevance backwards through layers
 - Constraint:
 - ◇ Total relevance is conserved at each layer
 - Result:
 - ◇ Each input receives a relevance score

3. Model-based methods

- LRP vs Gradient-based Methods
 - Gradients:
 - ◇ Measure sensitivity
 - LRP:
 - ◇ Decomposes the actual prediction
 - Key difference:
 - ◇ Gradients: *what would change the output?*
 - ◇ LRP: *what contributed to this output?*

From Attribution to Representation Analysis



University
of Exeter

- Previous methods:
 - Explain predictions via input contributions
- New perspective:
 - What does the model internally represent?
- Focus:
 - Internal mechanisms
 - Learned representations

Transformer Models and Attention



University
of Exeter

- Transformers rely on self-attention mechanisms
- Each token interacts with all others
- Attention computed using:
 - Queries, Keys, Values
- Produces:
 - Contextualized representations

Attention as an Interpretability Sign



University
of Exeter

- Attention provides:
 - A window into model processing
- Useful for:
 - Qualitative analysis
 - Exploring model behavior
- Widely used in tools:
 - Visualization frameworks (e.g., attention maps)

Important Caveat: What Attention Does NOT Guarantee

- Attention weights:
 - Do not necessarily reflect causal importance
- Observations:
 - Different attention patterns can lead to the same prediction
- Implication:
 - Attention alone is not a faithful explanation

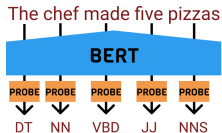
Probing

- Question:
 - What information is encoded in the model?
- Limitation of attention:
 - Does not directly measure stored knowledge
- Solution:
 - Probing tasks

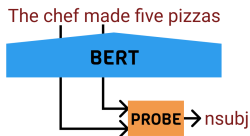
Probing Tasks: Core Idea

- Extract internal representations from the model
- Train simple classifiers on top of them
- Goal:
 - Test if specific information is encoded

Part-of-speech!



Partial dependency info!



What Do Probing Tasks Measure?

- Linguistic properties encoded in representations:
 - Syntax
 - Morphology
 - Coreference
- Interpretation:
 - If classifier performs well:
 - → information is present in representations

Interpretation of Probing Results

- Probing does NOT prove:
 - That the model uses this information
- Probing DOES show:
 - That the information is accessible
- Important distinction:
 - Encoding vs usage

Conclusion

- Covered content
- ILOs